

Virtual Hospitals Africa: Clinical Decision Support Architecture & Methodology

Virtual Hospitals Africa is a cloud-based patient management platform with embedded rules-based clinical decision support enabling health workers to more accurately prioritize and treat patients based on logical rules grounded national and international standards. [The interactive tutorial](#) gives a guided tour of the user experience from the perspective of a rural nurse conducting triage.

This document describes how that works: what guidance we utilize, how that is leveraged, the architecture behind the application and rules engine, and how we verify its accuracy.

Alignment with South African National Standards

The clinical rules in this system are drawn from South Africa's national standards:

South African Triage Scale (SATS) assigns every patient a priority level — **Emergency**, **Very Urgent**, **Urgent**, or **Non-urgent** — based on the presence of specific warning signs and a Triage Early Warning Score that is weighted higher for abnormal vital signs. In addition to informing the priority level and target time to treatment, the [SATS Manual](#) gives additional tasks and investigations to perform such as providing analgesia for burns or measuring blood oxygen levels if the respiratory rate is at all abnormal.

Practical Approach to Care Kit (PACK) / Adult Primary Care (APC) guidelines provide symptom-by-symptom clinical pathways for primary care. The [Adult Primary Care guidelines](#) provides a 1-2 page reference for common categories of patient complaints and finding sites (body structures). For each, the document recommends findings/measurements to check for that would be urgent if present and/or indicative of a specific underlying condition e.g., Patient has a lump -> check for a pulsating lump which would indicate that an aneurysm is likely and that the case should be referred urgently.

Standard Treatment Guidelines (STG) and Essential Medicines List (EML) inform the recommended doses and treatment protocols surfaced when a diagnosis is confirmed. Medicines are indexed against the South African EML, and recommended dose ranges are encoded with reference to the EML Version 1.1.

Architecture

Virtual Hospitals Africa is an open source Deno/Fresh web application deployed to Amazon Web Services's Cape Town data center, their only such hosting facility on the African continent. Clinical decision support is backed by a deterministic rules engine, described in more detail below. That rules engine reads and writes records based on an underlying medical model with several properties that enable flexibility, traceability, and flexibility.

Medical Model

Within VHA, patient records form an append-only graph with each record having a "root" [SNOMED](#) concept describing its category and a "specific" SNOMED concept describing the record. Records can evaluate or qualify other records, which may themselves be evaluated or qualified enabling records of arbitrary depth, e.g., a "Sudden Onset" record qualifying an "Itching" record means "Sudden Onset Itching". To edit a record we save a new record that evaluates the first record as "Entered in error".

There are many advantages of this approach over saving records as editable free text:

1. **Canonical representation:** health workers can search for medical concepts by their alias, but records will appear in the interface using the term's fully specified name for consistency
2. **Set relationships:** as we'll describe in more detail in reference to the rules engine, SNOMED's rich ontology encodes many relationships between concepts including finding sites, associated morphology, and "is a" relationships. So by using codes to represent records a health worker can record "Snake bite" and the a rule for "Bite — wound" will fire because a "Snake bite" "is a" "Bite — wound".
3. **Traceability & Improved Error Handling:** disallowing records from being modified makes the system more parallelizable and traceable, key characteristics if the system ever does not behave as inspected. Because we save records in perpetuity without modification, we can audit precisely how a record came to be and potentially remediate.

Rules Engine

The rules engine is a deterministic system where rules are encoded as S-expressions, taking effect when some condition has been met. So in the example,

```
(system_diagnosis_rule
  "Diagnose probable appendicitis based on nausea"
  (diagnosis
    (snomed_concept "Acute appendicitis" "disorder")
    probable
  )
  adult
  (and
    (or
      (clinical_finding (snomed_concept "Nausea" "finding"))
      (clinical_finding (snomed_concept "Finding of vomiting" "finding"))
    )
    (clinical_finding (snomed_concept "Tenderness of right lower quadrant of abdomen" "finding"))
  )
)
```

the system will diagnose that acute appendicitis is probable if there is an adult patient with tenderness of the right lower quadrant of their abdomen and either nausea or vomiting, as specified in the guidance in the APC 2023 manual.

- | | | |
|--|---|--|
| <ul style="list-style-type: none"> • Headache →30 • Chest pain →37 • If patient has watery diarrhoea (with or without vomiting) and has been in cholera outbreak area in past 5 days, cholera likely →47 • Neck stiffness, drowsy/confused or purple/red rash: meningitis likely <p>Manage and refer urgently:</p> <ul style="list-style-type: none"> • If BP < 90/60 or pancreatitis likely, give sodium chloride 0.9% 500mL IV over 30 minutes, repeat until systolic BP > 90. Continue 1L 6 hourly. Stop if breathing worsens. • If meningitis likely, give ceftriaxone 2g IV²/IM. Avoid injecting > 1g IM at one injection site. • If pain severe, give morphine 10mg IM or diluted morphine³ 3-10mg slow IV: start with 3mL IV over 3 minutes. If needed, give another 1mL/minute until pain improved, up to 10mL. Stop if BP drops < 90/60. • If glucose < 3 or ≥ 11.1 ↳ 17 or if diabetes and glucose < 4 ↳ 130. | <p>Give urgent attention to the patient with nausea/vomiting and any of:</p> <ul style="list-style-type: none"> • Guarding, rigidity or rebound tenderness: peritonitis likely • Tender in right lower abdomen: appendicitis likely • Sudden severe upper abdominal pain spreading to back: pancreatitis likely • BP < 90/60 • Vomiting blood | <ul style="list-style-type: none"> • Jaundice • Abdominal pain/distention and no stools or flatus/wind • Drowsy/confused/rapid deep breathing • If sudden nausea/vomiting and any of: generalised itch/rash, face/tongue swelling, wheeze, difficulty breathing, BP < 90/60, dizziness/collapse or exposure to possible allergen¹, check for anaphylaxis ↳ 20. |
|--|---|--|

The rules engine runs as part of an events-based system, processing patient records in between form submission and loading the subsequent page. Rules take the form of:

1. **Tasks & Investigations:** procedures for the health worker to do in response to findings, e.g., check for dilated pupils, measure blood oxygen saturation, raise the patient's legs. [Github](#)
2. **Reference Documents:** links to the specific pages of the reference materials allowing for cross-checking and more detailed visuals and instructions. [Github](#)
3. **Priority Evaluations:** evaluations of the patient's priority levels given a finding or diagnosis or a combination of findings. [Github](#)
4. **System Diagnoses:** evaluations of probable conditions based on a combination of findings. Corresponds with a condition being marked *likely* in the primary care guidelines. [Github](#)
5. **Recommended Dose Schedules:** recommended medications and dosages based on a confirmed diagnosis. Accounts for a patient's age, weight, and gives special instructions for health workers to check up on. [Github](#)

Evidence of Correctness

Several levels of testing help ensure that this clinical decision support system is accurate and performant.

Automated Testing

We run a suite [automated tests](#) that simulate a health worker taking a patient through the full triage workflow — from recording warning signs and measuring vital signs, to receiving a task list and a triage priority. These tests run against a real database and a running instance of the application; they do not use shortcuts or simulations.

The tests verify specific clinical scenarios drawn from the guidelines: for example, that a patient presenting with chest pain is prompted to check for an irregular pulse and severe pain; that a patient with an insect bite and low blood pressure is assigned a probable anaphylaxis diagnosis; and that the correct SATS urgency level is assigned based on the combination of recorded vital signs and symptoms. If any rule is changed in a way that alters clinical behaviour, these tests will fail, preventing that change from being deployed.

Load testing

In addition to testing the validity of the scenarios, we also measure the performance ensuring that latencies are low so that health workers can proceed smoothly through their work. Waiting on a slow system is painful in general and obviously that is even more the case in an urgent scenario, so we have taken several steps to ensure that processing is fast even for users on low range devices.

1. **Server-side processing:** almost all computation takes place on the server where we have more control and capacity compared with relying on users' devices

2. **Pre-compute:** A single rule may refer to a combination of concepts, each of which has many descendants. To latency runtime, we precompute for all SNOMED concepts what rules they could possibly apply to, turning a potentially long graph traversal into a simple lookup.
3. **Batch processing:** Generally speaking, a single form submission which may contain many findings is consolidated into a single request to the database
4. **Non-blocking logic:** We let the user record findings while rules are processing in the background for most pages

In load testing we found that the system performs well even with many concurrent patients, consistently processing the whole cycle (send completed web page, insert findings, run rules engine) in well under 400 milliseconds even with 100 patients being handled concurrently. With response times this fast, we expect that the rate-limiting factor to be the bandwidth of devices rather than the speed at which we generate responses. Contrast this with LLMs which for the higher reasoning models can have far longer runtimes. The results of these tests across different rules is included at the end of this document.

Manual code review by licensed medical professionals

Unlike black box language models whose behavior (with any nonzero [temperature](#)) is nondeterministic, our system has human-readable rules as artifacts.

Our Chief Medical Officer, Dr. Sikhululiwe Ngwenya, is leading the effort to cross-check all our rules for 1:1 parity with their corresponding guidance documents. If discrepancies are encountered now or in the future, the process to change them is simple, altering the rules and adding a test case to confirm the new behavior.

Manual user interface review by licensed medical professionals

The [apc-test-results](#) folder contains screenshots of the running application, one per presenting complaint, showing exactly what a health worker sees as the prioritization, reference documents, additional tasks during triage.



Warning Signs
Brief History
Height & Weight
Measure Vitals
Additional Tasks & Investigations
Assign Priority
Route Patient

Triage

Reference Documents

[APC 2023 — Fever](#)

Additional Investigations

Check For	Yes	No	Unknown
Due to Fever Diagnosis: Definite recorded at 5:21 PM			
Decreased level of consciousness*	☐	☐	☐
Stiff neck*	☐	☐	☐
Drowsy*	☐	☐	☐
Clouded consciousness*	☐	☐	☐
Petechiae of skin*	☐	☐	☐
Purpuric rash*	☐	☐	☐
Difficulty breathing*	☐	☐	☐
Tenderness of right lower quadrant of abdomen*	☐	☐	☐
Acute abdominal pain*	☐	☐	☐
Severe Backache*	☐	☐	☐
Jaundice*	☐	☐	☐
Easy bruising*	☐	☐	☐
Tendency to bleed*	☐	☐	☐
Seizure*	☐	☒	☐
Seizure: No recorded at 5:21 PM			
Check For	Yes	No	Unknown

TESTING

Check for **urgent fever** **Non-urgent conditions**

male • 1 January 1990

This Visit

Warning Signs
No findings entered

Brief History
No findings entered

Height & Weight
Height: 160 cm
Weight: 70 kg

Measure Vitals
Temperature: 39°C
Able to walk
Mentally alert
No traumatic injury
Respiratory rate: 12 bpm
Pulse: 60 bpm
Blood pressure: 120/80 mmHg

Additional Tasks & Investigation:
In progress

Diagnoses
Fever Diagnosis: Definite

History
Pre Existing Conditions
No findings entered

These screenshots serve as a visual audit trail: a clinician reviewing the guidelines can open the corresponding chapter of the APC document and verify that the system is asking for exactly the right things.

Conclusion

Virtual Hospitals Africa's architecture, testing, and review processes help ensure that our system and clinical guidance are accurate and responsive, ensuring that we can serve health workers even in resource poor settings.

num_patients	page	task definition	warning signs avg ms	warning signs median ms	brief history avg ms	brief history median ms	height and weight avg ms	height and weight median ms	measure vitals avg ms	measure vitals median ms
10	63-back-pain	Check for urgent back pain conditions	170	133	131	114	139	116	247	239
10	57-abnormal-vaginal-bleeding	Check for urgent vaginal bleeding conditions	143	145	114	114	109	108	216	212
10	28-collapse-falls	Check for urgent collapse conditions	178	170	102	101	105	104	237	238
10	26-weakness-or-tiredness	Check for urgent weakness/tiredness conditions	118	111	109	104	116	110	230	228
10	21-burns	Check for urgent burn conditions	140	135	106	101	109	107	241	241
10	21-burns	Measure Bloody Oxygen Saturation due to burns	146	143	107	107	108	106	242	233
10	19-seizures	Check for urgent seizure conditions	107	107	107	105	117	119	207	208
10	48-constipation-and-anal-symptoms	Check for urgent constipation conditions	106	106	112	112	128	126	214	208
10	48-constipation-and-anal-symptoms	Check for urgent anal conditions	115	104	116	118	130	127	211	207
10	37-chest-pain	Check for urgent chest pain conditions	226	171	115	107	134	122	235	222
10	41-acute-covid-19	Check for urgent acute Covid-19 conditions	114	109	112	110	121	120	264	256
10	80-scalp-symptoms	Check for urgent scalp conditions	111	112	116	115	140	136	220	217
10	36-gum-teeth-symptoms	Check for urgent dental conditions	112	109	114	114	122	115	256	258
10	79-changes-in-skin-colour	Check for urgent skin colour change conditions	136	111	126	115	148	137	290	275
10	59-urinary-symptoms	Check for urgent urinary symptom conditions	110	109	112	107	133	129	253	250
10	137-ischaeamic-heart-disease	Check for urgent ischaemic heart disease conditions	128	113	114	111	127	118	209	211

num_patients	page	task definition	warning signs avg ms	warning signs median ms	brief history avg ms	brief history median ms	height and weight avg ms	height and weight median ms	measure vitals avg ms	measure vitals median ms
10	42-ongoing-covid-19-symptoms	Check for urgent ongoing Covid-19 symptoms	108	108	106	106	123	122	236	236
10	39-wheeze-or-tight-chest	Check for urgent wheeze or tight chest conditions	127	120	153	113	118	109	208	209
10	82-nail-symptoms	Check for urgent nail symptom conditions	138	122	129	118	133	132	341	316
10	164-postnatal-care	Check for urgent postnatal conditions	115	110	119	115	136	129	242	216
10	139-peripheral-vascular-disease	Check for urgent peripheral vascular disease conditions	114	110	119	110	126	124	216	220
10	86-low-mood-stress-and-anxiety	Check for urgent low mood, stress and anxiety conditions	120	114	111	110	141	134	234	227
10	66-foot-symptoms	Check for urgent foot symptom conditions	132	117	117	115	138	129	284	285
10	73-drug-rash	Check for urgent drug rash conditions	129	126	115	113	118	117	256	265
10	45-nausea-or-vomiting	Check for urgent nausea or vomiting conditions	258	226	116	115	125	115	264	247
10	29-dizziness	Check for urgent dizziness conditions	190	185	104	103	111	110	194	191
10	25-lump-swelling-in-neck-axilla-or-groin	Check for urgent groin lump conditions	114	112	111	112	121	111	215	214
10	40-covid-19-diagnosis	Check for presence of Covid-19	107	103	107	105	121	122	211	210
10	65-leg-symptoms	Check for urgent leg symptom conditions	134	112	110	109	144	138	261	256
10	159-antenatal-care	Check for urgent antenatal conditions	114	112	132	117	127	131	225	226
10	44-abdominal-pain	Check for urgent abdominal pain conditions	192	177	103	103	116	104	220	215

num_patients	page	task definition	warning signs avg ms	warning signs median ms	brief history avg ms	brief history median ms	height and weight avg ms	height and weight median ms	measure vitals avg ms	measure vitals median ms
10	51-abnormal-vaginal-discharge	Check for urgent female genitalia conditions	110	107	123	118	132	133	224	217
10	33-ear-symptoms	Check for urgent ear symptom conditions	139	110	119	115	130	126	230	219
10	34-nose-symptoms	Check for urgent nose conditions	136	118	129	118	149	133	313	311
10	83-self-harm-or-suicide	Check for urgent self-harm or suicide conditions	126	121	123	114	125	120	249	251
10	31-eye-vision-symptoms	Check for urgent eye or vision conditions	118	113	114	109	123	120	264	264
10	78-crusts-or-flaky-skin	Check for urgent skin conditions with crusts or flaky skin	119	114	117	117	132	130	240	241
10	64-neck-pain-arm-hand	Check for urgent arm or hand symptom conditions	128	115	115	109	147	137	251	250
10	64-neck-pain-arm-hand	Check for urgent neck pain conditions	140	133	159	122	132	125	250	262
10	30-headache	Check for urgent headache conditions	143	139	123	122	132	125	296	292
10	32-face-symptoms	Check for urgent face symptom conditions	119	115	117	120	143	141	310	306
10	35-mouth-or-throat-symptoms	Check for urgent mouth or throat conditions	137	122	112	112	140	130	255	258
10	46-diarrhoea	Check for urgent diarrhoea conditions	136	121	121	112	114	108	235	228
10	38-cough-or-difficulty-breathing	Check for urgent cough/breathing conditions	214	214	137	122	117	117	223	207
10	22-bites-and-stings	Check for urgent bite/sting conditions	141	139	122	110	148	117	241	237
10	22-bites-and-stings	Check for snake bite	177	166	119	117	128	121	218	214
10	62-joint-symptoms	Check for urgent joint conditions	126	118	120	116	137	131	262	254

num_patients	page	task definition	warning signs avg ms	warning signs median ms	brief history avg ms	brief history median ms	height and weight avg ms	height and weight median ms	measure vitals avg ms	measure vitals median ms
10	67-skin-symptoms	Check for urgent skin symptom conditions	133	122	131	120	131	131	268	254
10	75-skin-ulcer-or-non-healing-wound	Check for urgent skin ulcer and wound conditions	129	121	114	115	155	126	295	287
10	74-skin-lumps	Check for urgent skin lump conditions	114	116	105	105	128	123	275	274
10	74-skin-lumps	Measure lesion size	121	119	112	112	130	133	272	270
10	50-genital-symptoms-man	Check for urgent genital symptoms in a man	135	118	112	114	136	127	232	224
100	63-back-pain	Check for urgent back pain conditions	137	131	115	112	120	110	244	236
100	57-abnormal-vaginal-bleeding	Check for urgent vaginal bleeding conditions	141	136	117	109	113	110	222	217
100	28-collapse-falls	Check for urgent collapse conditions	185	174	109	105	115	111	252	242
100	26-weakness-or-tiredness	Check for urgent weakness/tiredness conditions	120	111	113	108	122	118	241	240
100	21-burns	Check for urgent burn conditions	148	142	111	108	115	111	261	256
100	21-burns	Measure Bloody Oxygen Saturation due to burns	153	147	116	114	112	111	254	255
100	19-seizures	Check for urgent seizure conditions	123	113	116	114	129	128	223	217
100	48-constipation-and-anal-symptoms	Check for urgent constipation conditions	118	110	113	111	124	124	223	223
100	48-constipation-and-anal-symptoms	Check for urgent anal conditions	118	110	114	111	125	126	217	219
100	37-chest-pain	Check for urgent chest pain conditions	185	175	119	115	115	113	231	227
100	41-acute-covid-19	Check for urgent acute Covid-19 conditions	124	114	118	116	134	132	280	274

num_patients	page	task definition	warning signs avg ms	warning signs median ms	brief history avg ms	brief history median ms	height and weight avg ms	height and weight median ms	measure vitals avg ms	measure vitals median ms
100	80-scalp-symptoms	Check for urgent scalp conditions	124	113	117	115	131	128	236	227
100	36-gum-teeth-symptoms	Check for urgent dental conditions	125	116	116	116	136	131	274	272
100	79-changes-in-skin-colour	Check for urgent skin colour change conditions	134	119	120	118	140	138	291	286
100	59-urinary-symptoms	Check for urgent urinary symptom conditions	128	117	121	119	139	135	280	279
100	137- ischaemic- heart- disease	Check for urgent ischaemic heart disease conditions	130	121	126	120	125	121	232	233
100	42-ongoing-covid-19-symptoms	Check for urgent ongoing Covid-19 symptoms	126	116	117	115	145	135	259	254
100	39-wheeze-or-tight-chest	Check for urgent wheeze or tight chest conditions	141	129	121	118	128	120	237	242
100	82-nail-symptoms	Check for urgent nail symptom conditions	124	113	114	111	134	132	296	290
100	164-postnatal-care	Check for urgent postnatal conditions	121	113	119	118	134	132	228	226
100	139-peripheral-vascular-disease	Check for urgent peripheral vascular disease conditions	122	117	118	116	132	128	230	231
100	86-low-mood-stress-and-anxiety	Check for urgent low mood, stress and anxiety conditions	136	125	127	123	147	142	253	254
100	66-foot-symptoms	Check for urgent foot symptom conditions	129	117	122	119	145	141	303	299
100	73-drug-rash	Check for urgent drug rash conditions	137	130	125	120	125	121	277	275
100	45-nausea-or-vomiting	Check for urgent nausea or vomiting conditions	241	226	122	116	129	123	981	944
100	29-dizziness	Check for urgent dizziness conditions	202	193	126	125	123	119	225	218

num_patients	page	task definition	warning signs avg ms	warning signs median ms	brief history avg ms	brief history median ms	height and weight avg ms	height and weight median ms	measure vitals avg ms	measure vitals median ms
100	25-lump- swelling-in- neck-axilla- or-groin	Check for urgent groin lump conditions	127	121	135	130	141	141	244	242
100	40-covid-19- diagnosis	Check for presence of Covid-19	129	121	131	130	150	147	237	235
100	65-leg- symptoms	Check for urgent leg symptom conditions	126	120	124	122	143	143	257	251
100	159- antenatal- care	Check for urgent antenatal conditions	120	116	122	122	140	141	230	230
100	44- abdominal- pain	Check for urgent abdominal pain conditions	196	191	120	119	128	123	717	698
100	51- abnormal- vaginal- discharge	Check for urgent female genitalia conditions	120	116	124	123	142	143	222	223
100	33-ear- symptoms	Check for urgent ear symptom conditions	123	118	128	126	143	143	228	228
100	34-nose- symptoms	Check for urgent nose conditions	129	121	124	124	144	142	282	280
100	83-self- harm-or- suicide	Check for urgent self-harm or suicide conditions	124	121	124	123	143	145	237	237
100	31-eye- vision- symptoms	Check for urgent eye or vision conditions	124	120	121	121	142	144	261	257
100	78-crusts- or-flaky-skin	Check for urgent skin conditions with crusts or flaky skin	125	118	124	124	144	144	240	239
100	64-neck- pain-arm- hand	Check for urgent arm or hand symptom conditions	134	129	127	126	149	146	259	252
100	64-neck- pain-arm- hand	Check for urgent neck pain conditions	136	128	135	130	148	145	235	232
100	30- headache	Check for urgent headache conditions	148	139	126	125	133	129	260	256
100	32-face- symptoms	Check for urgent face symptom conditions	132	123	127	125	146	145	278	274
100	35-mouth- or-throat- symptoms	Check for urgent mouth or throat conditions	132	122	127	126	151	145	255	253

num_patients	page	task definition	warning signs avg ms	warning signs median ms	brief history avg ms	brief history median ms	height and weight avg ms	height and weight median ms	measure vitals avg ms	measure vitals median ms
100	46-diarrhoea	Check for urgent diarrhoea conditions	145	138	128	128	133	129	234	233
100	38-cough-or-difficulty-breathing	Check for urgent cough/breathing conditions	237	229	132	132	130	128	735	711
100	22-bites-and-stings	Check for urgent bite/sting conditions	158	155	128	128	130	126	242	241
100	22-bites-and-stings	Check for snake bite	161	157	135	133	126	124	246	245
100	62-joint-symptoms	Check for urgent joint conditions	137	128	134	131	152	150	250	249
100	67-skin-symptoms	Check for urgent skin symptom conditions	151	139	142	139	174	171	295	291
100	75-skin-ulcer-or-non-healing-wound	Check for urgent skin ulcer and wound conditions	144	136	137	135	172	166	313	305
100	74-skin-lumps	Check for urgent skin lump conditions	141	134	136	133	170	167	311	305
100	74-skin-lumps	Measure lesion size	135	132	137	134	169	167	311	306
100	50-genital-symptoms-man	Check for urgent genital symptoms in a man	156	140	147	141	182	171	289	268